Project 4-1: Even or Odd Checker

Create a program that checks whether a number is even or odd.

Console

```
Even or Odd Checker

Enter an integer:  33
This is an odd number.
```

Specifications

- Store the code that gets user input and displays output in the main function.
- Store the code that checks whether the number is even or odd in a separate function.
- Assume that the user will enter a valid integer.

Project 4-2: Hike Calculator

Create a program that converts the number of miles that you walked on a hike to the number of feet that you walked.

Console

```
Hike Calculator

How many miles did you walk?: 4.5
You walked 23760 feet.
```

Specifications

- The program should accept a float value for the number of miles.
- Store the code that gets user input and displays output in the main function.
- There are 5280 feet in a mile.
- Store the code that converts miles to feet in a separate function. This function should return an int value for the number of feet.
- Assume that the user will enter a valid number of miles.

Project 4-3: Feet and Meters Converter

Create a program that converts feet to meters and vice versa.

Console

```
Feet and Meters Converter

Conversions Menu:
a. Feet to Meters
b. Meters to Feet
Select a conversion (a/b): a

Enter feet: 100
30.48 meters

Would you like to perform another conversion? (y/n): y

Conversions Menu:
a. Feet to Meters
b. Meters to Feet
Select a conversion (a/b): b

Enter meters: 100
328.08 feet

Would you like to perform another conversion? (y/n): n

Thanks, bye!
```

Specifications

- The formula for converting feet to meters is:

  `feet = meters / 0.3048`

- The formula for converting meters to feet is:

  `meters = feet * 0.3048`

- Store the code that performs the conversions in functions within a module. For example, store the code that converts feet to meters in a function in a module.
- Store the code that displays the title in its own function, and store the code that displays the menu in its own function, but store the rest of the code that gets input and displays output in a main function.
- Assume the user will enter valid data.
- The program should round results to a maximum of two decimal places.

Project 4-5: Dice Roller

Create a program that uses a function to simulate the roll of a die.

Console

```
Dice Roller

Roll the dice? (y/n): y

Die 1: 3
Die 2: 6
Total: 9

Roll again? (y/n): y

Die 1: 1
Die 2: 1
Total: 2
Snake eyes!

Roll again? (y/n): y

Die 1: 6
Die 2: 6
Total: 12
Boxcars!

Roll again? (y/n): n
```

Specifications

- The program should roll two six-sided dice.
- Store the code that rolls a single die in a function.
- Store the code that gets input and displays output in the main function. Whenever it's helpful, use helper functions to split this code into other functions.
- The program should display a special message for two ones (snake eyes) and two sixes (boxcars).

Project 4-6: Prime Number Checker

Create a program that checks whether a number is a prime number and displays the total number of factors if it is not a prime number.

Console

```
Prime Number Checker

Please enter an integer between 1 and 5000: 1
Invalid integer. Please try again.
Please enter an integer between 1 and 5000: 2
2 is a prime number.

Try again? (y/n): y

Please enter an integer between 1 and 5000: 3
3 is a prime number.

Try again? (y/n): y

Please enter an integer between 1 and 5000: 4
4 is NOT a prime number.
It has 3 factors.

Try again? (y/n): y

Please enter an integer between 1 and 5000: 6
6 is NOT a prime number.
It has 4 factors.

Try again? (y/n): n

Bye!
```

Specifications

- A prime number is only divisible by two factors (1 and itself). For example, 7 is a prime number because it is only divisible by 1 and 7.
- If the number is not a prime number, the program should display its number of factors. For example, 6 has four factors (1, 2, 3, and 6).
- Store the code that gets a valid integer for this program in its own function.
- Store the code that calculates the number of factors for a number in its own function.
- Store the rest of the code that gets input and displays output in the main function.